

Python Tip: `sorted()`

Need to sort a list?

Need to sort it a specific way?

Use `sorted()` with the `key` argument to choose the **how** to sort it.



Basic Sorting

`sorted()` returns a new list with the values in order.

```
scores = [88, 73, 95, 82]

sorted_scores = sorted(scores)

print("Original:", scores)
print("Sorted:", sorted_scores)
```

```
Original: [88, 73, 95, 82]
Sorted: [73, 82, 88, 95]
```

The original list stays unchanged.

Sort Words by Length

By default, strings are sorted alphabetically. Use `key=` to choose a different thing to sort by, like the length of each string.

```
words = ["pear", "fig", "watermelon"]

alphabetical = sorted(words)
by_length = sorted(words, key=len) # Sort by length

print("Alphabetical:", alphabetical)
print("By length:", by_length)
```

```
Alphabetical: ['fig', 'pear', 'watermelon']
By length: ['fig', 'pear', 'watermelon']
```


Sort Without Case Sensitivity

Capital letters normally sort before lowercase letters. Use `str.lower` to compare lowercase versions of each word!

```
names = ["grace", "Ada", "linus", "Guido"]  
  
default = sorted(names)  
ignore_case = sorted(names, key=str.lower)  
  
print("Default:", default)  
print("Ignore case:", ignore_case)
```

```
Default: ['Ada', 'Guido', 'grace', 'linus']  
Ignore case: ['Ada', 'grace', 'Guido', 'linus']
```

Sort Dictionaries

Use a `lambda` function to choose the thing to sort by, like a value from a dictionary.

```
people = [  
    {"name": "Ada", "age": 36},  
    {"name": "Guido", "age": 31},  
    {"name": "Grace", "age": 28},  
]  
  
by_age = sorted(people, key=lambda p: p["age"])  
  
for p in by_age:  
    print(p["name"], p["age"])
```

```
Grace 28  
Guido 31  
Ada 36
```

Reverse the Order

Add `reverse=True` when you want the largest value first.

```
scores = [88, 73, 95, 82]

highest_first = sorted(scores, reverse=True)

print(highest_first)
```

```
[95, 88, 82, 73]
```

You can combine `reverse=True` with `key=` too.

Wrap-Up

Now you can customize how Python sorts lists!

Use `sorted()` to create a new sorted list.

Use `key=` to choose how Python sorts your list.

Add `reverse=True` when you want descending order.

Follow me for more tips.

Shep Bryan IV